Durethan® BCF30XH2.0

PA6-(GF+CF)30 Envalior

Injection Molding, 30% Glass Fiber/Carbon Fiber Reinforced, Heat Stabilized, Electrical Conductive

ISO 1043 PA6-(GF+CF)30

Rheological properties	dry / cond	Unit	Test Standard
ISO Data			
Melt volume-flow rate, MVR	15 / *	cm³/10min	ISO 1133
Temperature	260 / *	°C	-
Load	5 / *	kg	-
Molding shrinkage, parallel	0.2 / *	%	ISO 294-4, 2577
Molding shrinkage, normal	0.8 / *	%	ISO 294-4, 2577

Mechanical Properties	dry / cond	Unit	Test Standard
ISO Data			
Tensile Modulus	13500 / 6700	MPa	ISO 527
Stress at Break	195 / 100	MPa	ISO 527
Strain at Break	3.1 / 5	%	ISO 527
Impact Strength (Charpy), +23°C	75 / 80	kJ/m²	ISO 179/1eU
Impact Strength (Charpy), -30°C	60 / 60	kJ/m²	ISO 179/1eU
Notched Impact Strength (Charpy), +23°C	11 / -	kJ/m²	ISO 179/1eA
Puncture energy, +23°C	2.4 / -	J	ISO 6603-2
Puncture energy, -30°C	1.7 / -	J	ISO 6603-2

Thermal Properties	dry / cond	Unit	Test Standard
ISO Data			
Melting Temperature (10°C/min)	220 / *	°C	ISO 11357-1/-3
Temp. of deflection under load (1.80 MPa)	205 / *	°C	ISO 75-1/-2
Temp. of deflection under load (0.45 MPa)	219 / *	°C	ISO 75-1/-2
Coeff. of Linear Therm. Expansion, parallel	20 / *	E-6/K	ISO 11359-1/-2
Coeff. of Linear Therm. Expansion, normal	90 / *	E-6/K	ISO 11359-1/-2
Burning Behav. at thickness h	HB / *	class	UL 94
Thickness tested	3.0 / *	mm	-
UL recognition	yes / *	-	-

Electrical Properties	dry / cond	Unit	Test Standard
ISO Data			
Volume Resistivity	2000 / -	Ohm*m	IEC 62631-3-1

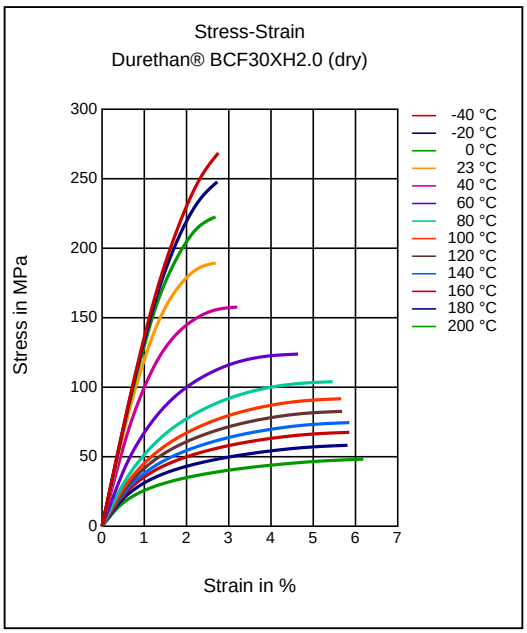
Other Properties	dry / cond	Unit	Test Standard
ISO Data			
Density	1330 / -	kg/m³	ISO 1183

Test specimen production	Value	Unit	Test Standard
ISO Data			
Injection Molding, melt temperature	280	°C	ISO 294
Injection Molding, mold temperature	80	°C	ISO 294

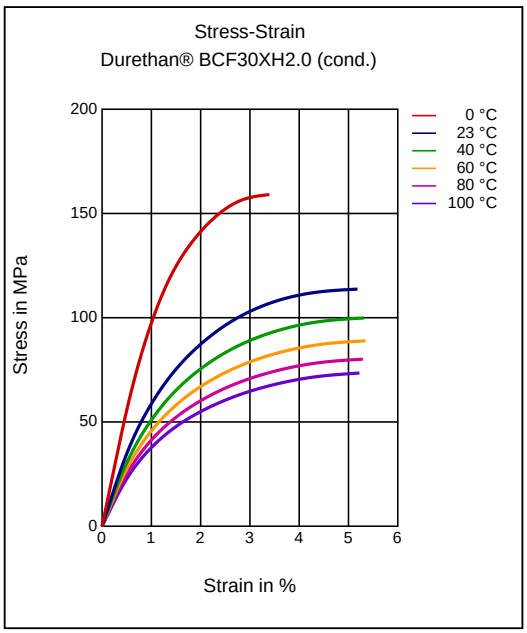
Processing Recommendation Injection Molding	Value	Unit	Test Standard
Pre-drying - Temperature	80	°C	-
Pre-drying - Time	2 - 6	h	-
Processing humidity	≤0.12	%	-
Melt temperature	270 - 290	°C	-
Mold temperature	80 - 120	°C	-

Diagrams

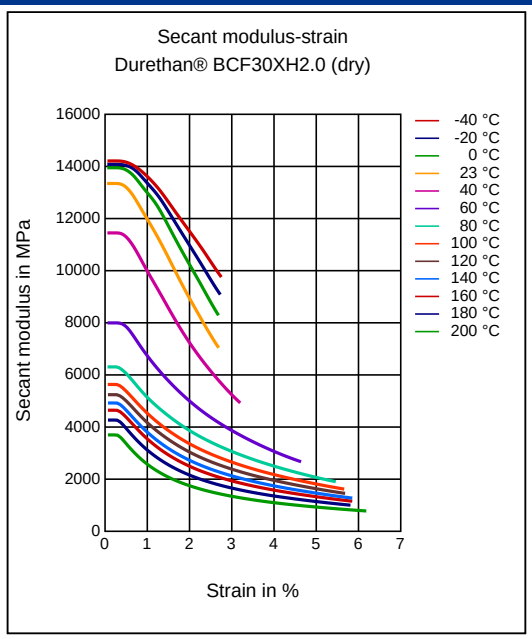
Stress-strain



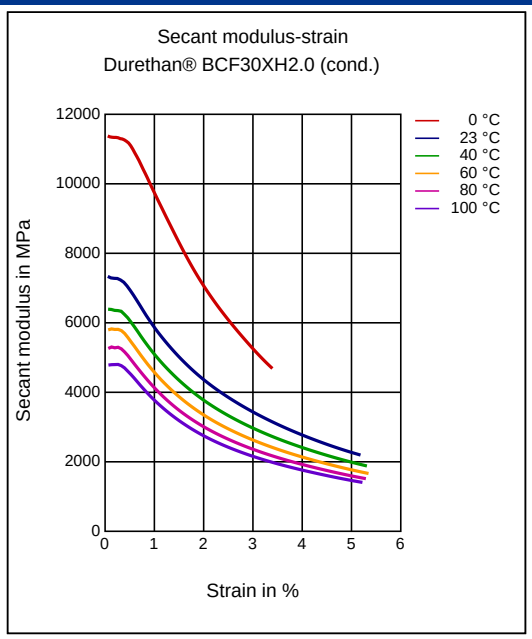
Stress-strain



Secant modulus-strain



Secant modulus-strain



Characteristics

Processing
Injection Molding

Special Characteristics
Electrically Conductive, Heat aging stabilized

Delivery form
Pellets

Injection Molding
PREPROCESSING

Residual moisture content: 0.03 - 0.12%
Drying temperature dry air dryer: 80 °C

Drying time dry air dryer 2 - 6 h

PROCESSING
Melt temperature (Tmin - Tmax): 270 - 290 °C

Mold temperature: 80 - 120 °C

Disclaimer

Liability Exclusion
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